

component that produces it (unless you are a biomechanics specialist). Rather, it is knowing what training will improve it.

Third, there is the force developed by the legs in pushing off for the first move to the ball. Whatever that force is for an individual player, it can be maximised by achieving a knee bend of roughly 90° for maximum push-off force. The degrees of knee bend for maximum push-off vary from player to player. It should be identified for each player by testing various degrees of knee bend.

### **What is the connection between a correct ready position and a properly executed split step?**

If a player has time to get into an ideal ready position, they should do so. This ready position should include a knee bend as close to 90° as possible. Having the proper arm placement and wrist bend will facilitate the unit turn just as the after-landing knee flex occurs.

It is possible — and becoming increasingly necessary — to initiate the split step hop from a position that is not the typical ready position. For example, a player sprints laterally, diagonally, or forward, and as they run, hop into a split step and rotate their body so as to touch down facing the net.

### **Based on your research, why has the split step been poorly written about, poorly taught, and poorly performed so far in tennis?**

I can't say for sure. One explanation might be that because a split step occurs so quickly, coaches and players might have casually assumed that it was instantaneous. Compared to the speed of balls in the early days of tennis, this might have been an inconsequential assumption. It certainly is not now. Split steps are not instantaneous. They have parts: 1) the hop, 2) the touchdown, 3) the after-landing knee bend (loading). Parts 1 and 2 take tenths of a second.

Another explanation might be that because the hop is the part of the split step that requires the most exertion, the split step was thought of as the hop. So, obviously, the

instruction would be to split step (hop) before the opponent hits the ball. A third explanation, probably now the most common, is that being airborne when the opponent hits the ball allows a player to detect the ball's direction and to land more forcefully on the foot they will push off from for their first move to the ball. Doing so is believed to store more energy in the push-off leg than if the player landed simultaneously on both feet. If more energy were stored, that extra push-off force is believed to produce a more rapid move to the oncoming ball.

### **But does this extra push-off force produce a faster move towards the oncoming ball? And if so, how relevant is this benefit?**

This extra push-off is questionable, but that is not the relevant question. The relevant question is: will being airborne and landing more forcefully on the push-off foot allow a player to reach the oncoming ball sooner after the opponent hits it? Answering this question must account for the fact that a player's first move to the ball is later than if they had touched down earlier.

Another consideration is how extra force is produced. There are two possibilities. One, which is a popular belief, is that landing on one leg stores elastic energy, and that energy is responsible for the extra force. Studies of the amount of extra force, measured as increased jump height of vertical jumps after a pre-jump-induced knee bend, only increased jump height by 5 per cent to 10 per cent. The effect on horizontal lunges, common in tennis, is probably less because the lunge movement is not as well aligned with the legs and hips as it is for a vertical jump.

The second belief rests on the conclusion by Van Hooren and Zolotarjova in a 2017 review of 93 relevant, peer-reviewed papers. They concluded that whatever extra force is created by landing after a pre-jump (a split step is a pre-jump) is primarily a function of knee bend and muscle activation rather than elastic energy, which has little effect. Therefore, the

increase in first-move speed would presumably be the same whether the landing was on one foot or two. And because the direction of the ball could be determined during the after-landing knee flex, the first move could begin 0.2 seconds after the opponent hit the ball.

The first belief underlies the belief that landing on the push-off foot results in a push-off towards the oncoming ball that is forceful enough to make up for the delay caused by the need to be airborne as the ball approaches.

The second belief would support three changes in split steps:

- ❶ The touchdown should occur at the instant the opponent hits the ball.
- ❷ The knees should be bent during the split step flight.
- ❸ The sum of the knee bend during flight and the after-landing knee bend would be roughly 90°, or whatever knee bend for maximum push-off force was unique to the player.

These are the important characteristics of the ideal split step.

### **What methods did you use to observe and measure splits on the pro tours?**

For the timing, I recorded slow-motion videos of televised tennis matches or watched *YouTube* videos at 1/4 speed, often also recorded on the iPhone in Slo-Mo mode to produce videos at 1/16 speed. I measured times with a stopwatch from these videos and corrected the times for the slower speed of those videos. I created stop-action photos from some of these slow-motion videos. These photos typically showed opponents' and players' ball contact, split-step hop initiation, and the timing of split-step touchdown.

### **What are the most common and serious split-step flaws on the pro tours?**

Without a doubt, the most serious flaw is landing late. Other problems arise from that flaw. As the game speeds up, time is the key feature. Landing late wastes valuable time.

The 'after-landing two-step' is another flaw. Players push their